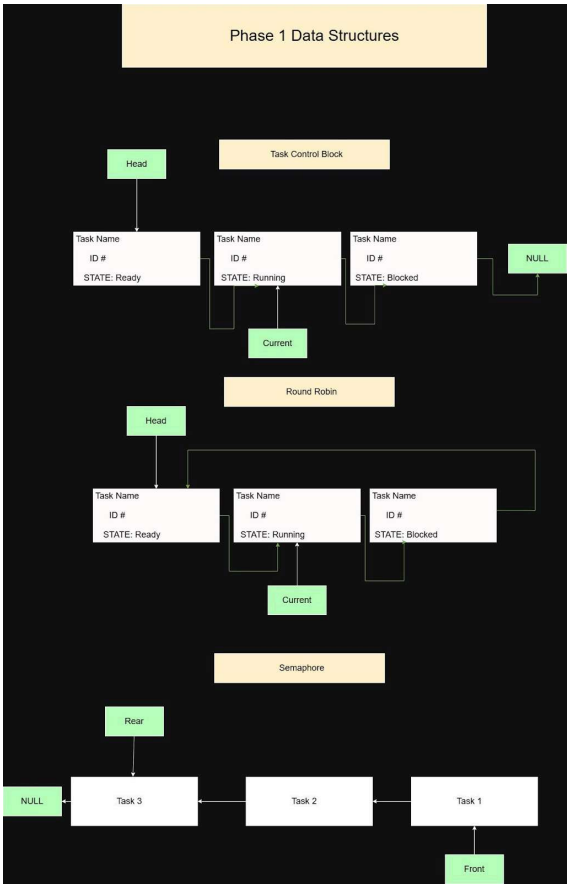
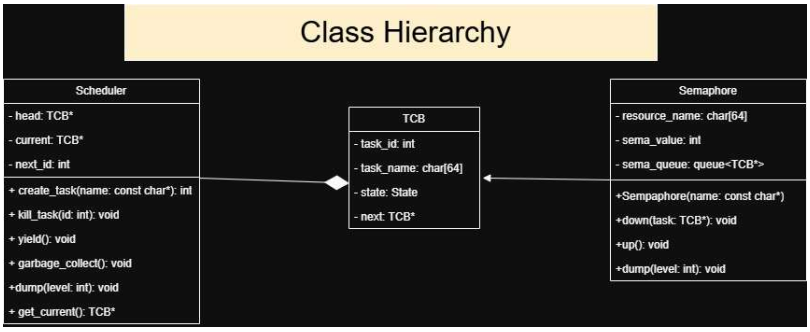
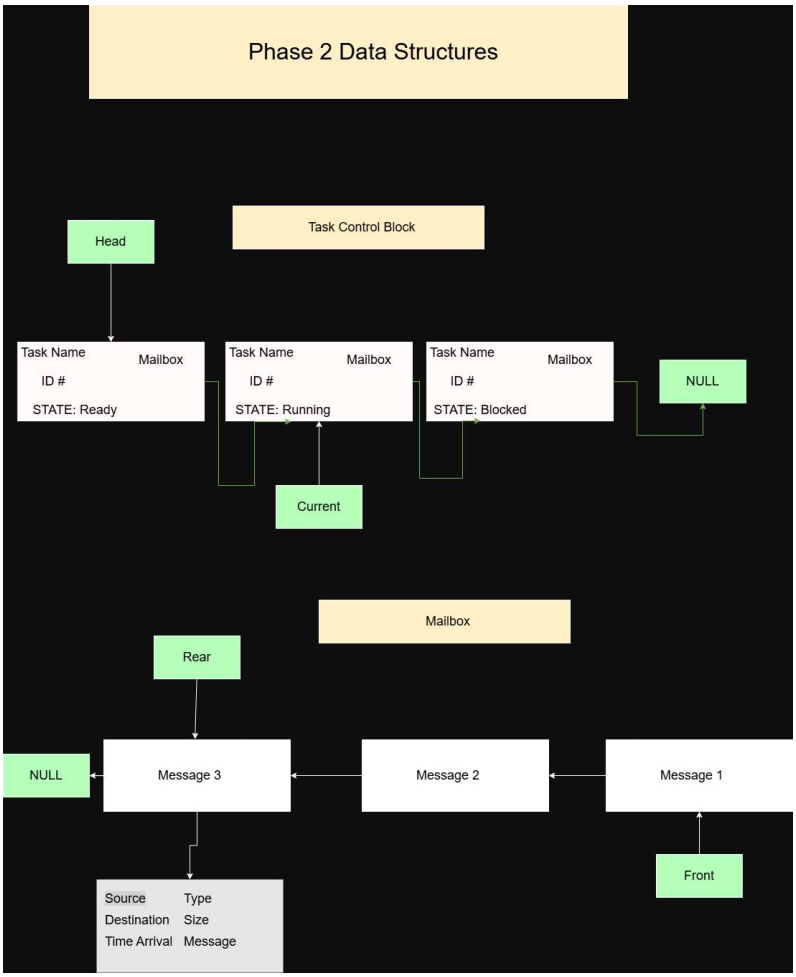
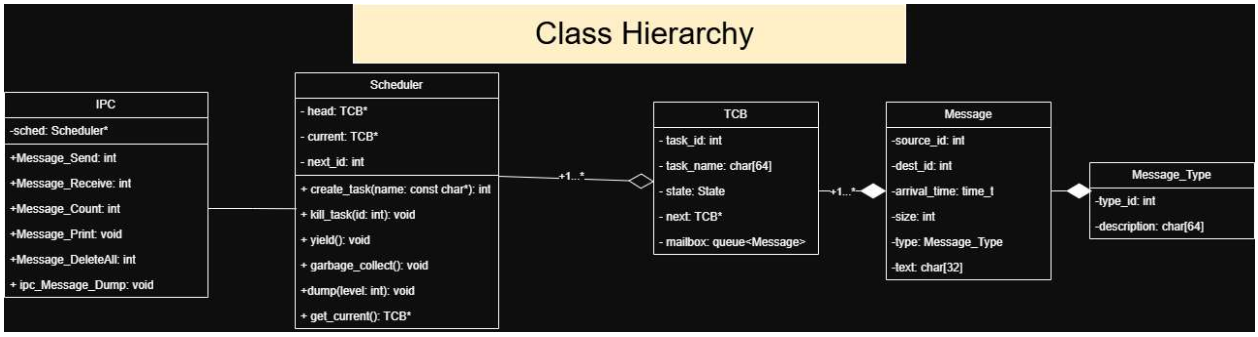
Design Diagrams:

A single linked list of TCBs is used to manage tasks within the scheduler. A pointer is in each TCB, pointing to the next task. This allows the scheduler to move, add, and delete tasks. There is a head pointer at the head of the list and a current pointer to the currently executing task, allowing round-robin scheduling. The semaphore uses a queue to block tasks when a shared resource is unavailable. Blocked tasks are stored in FIFO order. The scheduler and TCB have a composition relationship, as the TCB depends on the scheduler for creating and managing the tasks. The semaphore and TCB have an association relationship, as the semaphore references the task, using the queue to control access.





The message system uses a Task Control Block, which stores a task's name, ID, state, and a mailbox. TCB are linked together with a pointer at the head and one at the next task. Each TCB has a mailbox with a FIFO queue. Messages have a source, destination, message text, arrival time, and message size. The IPC class has an association with the Scheduler, as ICP uses the Scheduler for finding tasks. THe Scheduler has an aggregation relationship with the TCB. THe TCBs has a composition relationship with the Message objects as the mailbox is controlled by the TCB, and each message has a composition relationship with Message_Type.